PROJECT LEE — COGNITIVE RUNTIME + LONG-TERM MEMORY ARCHITECTURE

Yes. I would keep the entire cognition design you pasted and add a Memory Architecture Foundation underneath it, because practically every one of those cognitive abilities depends on LEE being able to retain years of experience without turning the Brain into an enormous, polluted, expensive database.

I would add the following to the design.

The cognitive objective remains:

A continuously running cognitive environment that maintains a model of Jesse, itself, Lamont Labs, relationships, projects, goals and the outside world; notices change; learns from outcomes; allocates attention; reasons about uncertainty; performs bounded work; and preserves its experience over years.

LEE should pursue extremely advanced functional cognition and operational awareness without making unsupported claims of consciousness or subjective experience.

The existing 20 cognitive capabilities remain:

1. Persistent situational awareness
2. Temporal awareness
3. Attention
4. Owner model
5. Self-model
6. Metacognition
7. Prediction
8. Counterfactual reasoning
9. Goal awareness
10. Initiative
11. Long-running thought
12. Relationship cognition
13. Cross-system causal reasoning
14. Learning from outcomes
15. Cognitive specialization
16. Internal reflection cycles
17. Curiosity / uncertainty-driven information seeking
18. Governed self-improvement
19. Multiple time horizons
20. Persistent cognitive runtime

The following memory architecture should now be treated as the foundation supporting all twenty.

21. MEMORY MUST BE TIERED, NOT MONOLITHIC

LEE must never treat “memory” as one giant prompt, one huge table, or one undifferentiated vector store.

Separate:

Durable Memory

Everything LEE has legitimately retained.

from:

Active Cognitive Context

The small amount of information required for the reasoning process currently taking place.

LEE may eventually contain millions of records and years of operational history while a particular reasoning task retrieves only a few dozen relevant records.

The intended flow is:

Millions of retained memories
→ retrieval and relevance filtering
→ bounded Context Packet
→ CIL
→ reasoning

LEE must never attempt to compensate for poor retrieval by stuffing her entire history into model context.

22. WORKING MEMORY

Create a deliberately small, fast Working Memory representing what LEE is actively thinking about.

Working Memory may include:

- current conversation
- active objective
- active project
- active person/relationship
- current investigation
- currently relevant evidence
- unresolved question
- active approvals
- recent important changes
- current operational state
- current plan
- current task
- recent reasoning conclusions

Working Memory is temporary and dynamic.

Information should enter and leave Working Memory based on attention and current objectives.

Leaving Working Memory must not imply deletion from durable memory.

23. STRUCTURED BRAIN MEMORY

PostgreSQL should remain the canonical structured Brain.

It should store durable structured concepts such as:

- People
- Organizations
- Projects
- Systems
- Objectives
- Initiatives
- Facts
- Interpretations
- Assumptions
- Decisions
- Commitments
- Waiting loops
- Risks
- Strategic Anchors
- Relationships
- Knowledge gaps
- Predictions
- Investigations
- Lessons
- Patterns
- Institutional Knowledge

- Brain versions
- operational state
- connection state
- permissions
- provenance
- reasoning telemetry
- economics records

This is the semantic skeleton of LEE's understanding.

Large binary files should not be dumped into PostgreSQL simply because PostgreSQL is the Brain database.

24. EPISODIC MEMORY

LEE should maintain an autobiographical history through the Event Log and related temporal structures.

Episodic Memory answers:

- What happened?
- When?
- In what order?
- What changed afterward?
- What was LEE doing?
- What was Jesse doing?
- What decision existed at that time?
- What did LEE believe then?
- What later evidence changed that belief?

Examples:

- QuantraCore CI failed.
- LEE detected it.
- LEE investigated it.
- LEE proposed a repair.
- Tests passed.
- Jesse approved it.
- The project returned to healthy state.

That sequence should remain reconstructable years later.

Older episodic memory may become colder or more compressed for retrieval purposes, but the underlying historical record must remain intact according to retention policy.

25. SOURCE EVIDENCE ARCHIVE

Large source material should live in a dedicated content-addressed archive rather than bloating the structured Brain.

Examples:

- PDFs
- Word documents
- spreadsheets
- images
- screenshots
- photographs
- audio
- video
- email attachments
- repository archives
- large exports
- generated artifacts
- raw connector packages

Each archived object should have a stable identity based on strong content hashing where appropriate.

The Brain stores things such as:

- Source ID
- hash
- content type
- source/provider
- original location
- archive location
- timestamps
- size
- owner/project/person relationships
- extracted text reference
- provenance
- integrity state
- retention state

LEE should therefore know about the artifact without carrying the artifact itself through every cognitive operation.

26. SEMANTIC INDEX

LEE should maintain a local semantic retrieval index over useful textual memory.

Possible indexed material:

- Evidence
- Emails
- Documents
- Facts
- Interpretations
- Decisions
- Commitments
- Lessons
- Institutional Knowledge
- Prior investigations
- Important project events
- Relationship history

The semantic index is not canonical memory.

It is an index into canonical memory.

If it is corrupted or deleted, LEE should be able to reconstruct it from canonical records.

That means:

Vector/index state must never become the sole surviving copy of an important memory.

Local embeddings should remain the preferred architecture where practical.

27. CONSOLIDATED LONG-TERM MEMORY

LEE should not preserve every observation at identical cognitive weight forever.

Memory should consolidate.

The intended progression is:

Raw Event
→ Experience
→ Lesson
→ Pattern
→ Institutional Knowledge

Example:

LEE experiences several dependency-related CI failures across projects.

Those individual events remain historically available.

Eventually LEE may learn:

Shared dependency upgrades have repeatedly caused cross-project CI failures. Verify dependency alignment before repairing individual downstream projects.

That consolidated memory becomes easier to retrieve than dozens of raw incidents.

But it must retain provenance back to the supporting experiences.

28. MEMORY TEMPERATURE

Give memory a retrieval-temperature model.

HOT

Frequently needed or immediately relevant.

Examples:

- current projects
- active objectives
- recent important correspondence
- current commitments
- recent project activity
- current system state
- unresolved decisions

Optimized for fast retrieval.

WARM

Still useful but not continuously active.

Examples:

- earlier stages of current projects
- previous decisions
- older correspondence
- completed initiatives
- historical relationship context

Searchable without remaining in the hottest indexes.

COLD

Rarely needed but still worth preserving.

Examples:

- old project artifacts
- historical attachments
- old raw events
- closed investigations
- long-completed initiatives

May be compressed or moved to archival storage.

CANONICAL / PERMANENT

Some information must not become unimportant merely because it is old.

Examples:

- Constitutional rules
- owner-declared identity
- Strategic Anchors
- critical historical decisions

- important legal/commercial evidence
- institutional knowledge
- evidence required for audit
- durable owner corrections

Temperature and importance must remain separate concepts.

A ten-year-old Strategic Anchor may be more important than yesterday's email.

29. MEMORY IMPORTANCE

Each durable memory should be able to carry dimensions such as:

- importance
- freshness
- confidence
- provenance strength
- usage frequency
- current-objective relevance
- project relevance
- relationship relevance
- operational impact
- contradiction status
- canonical status
- retention class

Do not collapse all of these into one opaque score.

LEE should be able to explain why something is prioritized.

30. MEMORY CONSOLIDATION CYCLES

Expand LEE's reflection cycles into explicit memory consolidation.

A nightly or scheduled cognitive-maintenance cycle could perform:

OBSERVE

What happened?

RECONCILE

Did new evidence contradict existing records?

DEDUPLICATE

Are multiple records describing the same event or source?

REVIEW

Which assumptions or interpretations became stale?

LEARN

Did a decision produce an observable outcome?

CONSOLIDATE

Can repeated experiences form a useful lesson or pattern?

COMPRESS

Can verbose historical material be represented by a higher-level memory while retaining provenance?

COOL

Which memories no longer need hot storage/index priority?

ARCHIVE

Which large artifacts should move to archival storage?

REINDEX

Which semantic indexes require refresh?

PREPARE

What should be placed into tomorrow's operational briefing?

The cycle should improve organization, not rewrite history.

31. MEMORY MUST PRESERVE HISTORICAL BELIEF STATE

LEE must be able to distinguish:

What was true

from:

What LEE believed was true at a particular point in time.

Example:

In 2026 LEE believed a pilot was likely to begin soon.

Later evidence showed the timeline was much slower.

LEE must not rewrite the old record to pretend she always knew the later outcome.

Instead:

Interpretation at T1
→ later evidence
→ revised interpretation at T2

That gives LEE genuine temporal epistemology.

She can learn from being wrong.

32. MEMORY POLLUTION PROTECTION

Bad memory is more dangerous than insufficient storage.

LEE must never automatically turn model output into fact.

Maintain strict separation between:

- Evidence
- Facts
- Interpretations
- Assumptions
- Predictions
- Owner declarations
- Lessons
- Institutional Knowledge

An interpretation becoming frequently repeated does not make it a fact.

A model agreeing with itself does not count as independent evidence.

A prediction coming true once does not automatically become a durable heuristic.

This should be treated as a constitutional memory rule.

33. OWNER MODEL MEMORY

LEE's model of Jesse should become its own governed memory domain.

It can contain:

Explicit Owner Truth

Things Jesse directly said.

Observed Behavior

Things supported by recorded actions.

Inferred Preference

Patterns LEE believes may exist.

Decision Heuristic

Repeated decision behavior.

Stable Strategic Anchor

Explicit durable guidance.

Uncertain Owner Hypothesis

A pattern LEE suspects but has insufficient evidence to rely on strongly.

Every owner-model record must clearly identify which category it belongs to.

LEE must never present inferred traits as if Jesse explicitly declared them.

34. SELF MEMORY

LEE should preserve meaningful history about herself.

Examples:

- version history
- capability additions
- capability removals
- failures
- outages
- repairs
- performance
- prediction accuracy
- recommendation outcomes
- project-operation outcomes
- connector reliability
- cost history
- previous operational-confidence states
- self-improvement proposals
- rejected self-improvement proposals

This gives the Self Model continuity.

LEE should know not only:

What can I do?

but eventually:

What could I do six months ago?

and:

Which parts of myself have repeatedly failed?

35. RELATIONSHIP MEMORY

Each important relationship should accumulate longitudinal history.

Not just messages.

LEE should retain:

- communication events
- important topics
- shared projects
- commitments
- unresolved questions
- decisions
- normal response cadence
- major deviations
- meetings
- important documents
- previous successful interaction patterns
- boundaries
- known communication preferences

This supports sophisticated Relationship Cognition while avoiding unjustified claims about another person's private mental state.

36. PROJECT MEMORY

Projects should develop their own memory streams.

Each project may retain:

- origin
- objectives
- major architecture changes
- decisions
- rejected directions
- important failures
- repairs
- deployments
- integrations
- contributors
- milestones
- lessons
- current assumptions
- unresolved technical debt
- dependencies
- historical momentum
- current state

Then LEE can answer:

Why is QuantraCore built this way?

instead of only:

What files currently exist?

37. INVESTIGATION MEMORY

Long-running thought requires persistent Investigation objects.

An Investigation should retain:

- question
- objective
- created date
- current hypothesis
- supporting evidence
- contradictory evidence
- unresolved questions
- assumptions
- research performed
- related people/projects
- conclusions
- confidence
- conclusion history
- current status
- next evidence needed

Investigations can remain open for days, weeks, months, or years.

New evidence can automatically trigger reconsideration.

This allows cognition to continue beyond a chat session.

38. PREDICTION MEMORY

Predictions should be stored separately from facts.

Every meaningful prediction should retain:

- prediction
- timestamp
- predicted time horizon
- supporting evidence
- reasoning
- confidence range
- relevant variables
- eventual outcome
- accuracy result
- lesson derived

LEE should periodically score her predictions.

This creates real calibration over time.

Eventually she can know:

I am historically reliable at predicting project delay but poorly calibrated on human response timing.

That is useful metacognition.

39. DECISION → OUTCOME MEMORY

Decisions should not end when they are recorded.

Build an explicit lifecycle:

Decision
→ Expected outcome
→ Observed outcome
→ Difference
→ Lesson
→ Potential heuristic

Example:

Decision: Wait before following up.

Expected: Allow the relationship time without unnecessary pressure.

Outcome: Reply arrived nine days later without follow-up.

Lesson: Waiting was successful under this context.

Repeated similar outcomes can eventually inform Decision Memory.

This is how LEE learns from your actual life.

40. CAUSAL MEMORY

LEE should preserve causal relationships distinctly from simple correlation.

Example:

API contract changed → caused QuantraCore integration test failure → caused Webull task delay → affected today's priority ordering

Causal claims should carry:

- evidence
- confidence
- source
- whether explicitly known or inferred
- alternative explanations if significant

LEE must not casually transform sequence into causation.

41. KNOWLEDGE GAPS AS MEMORY

Curiosity requires LEE to remember what she does not know.

Create durable Knowledge Gap objects.

Examples:

- uncertain project status
- contradictory pilot timeline
- unknown fulfillment of commitment
- missing deployment evidence
- unavailable account state
- unresolved person identity

Knowledge gaps can have:

- importance
- reason
- affected objective
- possible evidence sources
- last investigation date
- next allowed action

LEE can then:

- search existing evidence
- ask Jesse
- inspect a project
- query a provider
- wait for an external event
- leave the uncertainty unresolved

Not knowing should be a first-class state.

42. MEMORY GARBAGE COLLECTION

LEE needs maintenance without destructive forgetting.

A Memory Garbage Collector can remove or compact things such as:

- duplicate imported files
- duplicate provider payloads
- redundant embeddings
- obsolete caches
- temporary extraction files
- expired diagnostic artifacts
- temporary project clones
- orphaned transient records
- rebuildable indexes
- superseded temporary representations

It must not automatically delete:

- canonical evidence
- Event Log history
- Strategic Anchors
- constitutional history
- owner declarations
- important decisions
- audit evidence
- institutional knowledge

Deletion policy must remain explicit and auditable.

43. SOURCE-SPECIFIC RETENTION

Different providers require different storage strategies.

Gmail

Gmail remains the external canonical mailbox.

LEE may retain:

- Gmail message/thread IDs
- metadata
- important normalized body content
- entities
- commitments
- relationships
- evidence references
- selected snapshots where preservation matters

LEE does not necessarily need to duplicate every raw Gmail byte forever.

GitHub

GitHub/Git history remains canonical for repository history.

LEE retains:

- repo identity
- commit references
- meaningful diffs
- build state
- project relationships
- important change summaries
- operational evidence

LEE should not necessarily maintain endless redundant full repository copies.

Drive

Drive remains canonical for cloud files while connected.

LEE retains:

- file identity
- version metadata
- relationships
- extracted evidence
- important snapshots where needed

Local captures

LEE may become the canonical source for owner-created captures and therefore requires stronger preservation guarantees.

Retention policy should understand source ownership.

44. MEDIA STORAGE

Images, audio and video will eventually dominate storage much more than text.

Use specialized archival storage.

For example:

Voice capture:

Original audio
→ archive according to retention policy

Transcript
→ evidence

Extracted facts / commitments
→ structured Brain

Embedding
→ semantic index

Summary
→ cognitive memory if useful

This prevents heavy media from becoming the active Brain.

45. STORAGE EXPANSION MUST BE TRANSPARENT TO LEE

Do not design LEE around a permanently fixed K6 disk size.

Storage should be expandable.

Eventually the preferred physical layout could be:

FAST INTERNAL STORAGE

- PostgreSQL
- active Brain
- semantic indexes
- hot evidence
- working data
- Event Log hot partitions

LARGE ENCRYPTED ARCHIVE

- historical sources
- screenshots
- attachments
- audio
- video

- cold evidence
- historical exports

INDEPENDENT BACKUP DESTINATION

- Brain backups
- database backups
- Event Log recovery
- archive manifest
- integrity metadata

LEE should understand storage tiers without changing her conceptual memory model.

46. MEMORY PORTABILITY

LEE's accumulated cognition must outlive the machine.

Create a portable Brain export/restore architecture containing or referencing:

- canonical database
- Event Log
- Brain versions
- source manifest
- archive manifest
- institutional knowledge
- relationship graph
- project memory
- owner model
- self model
- investigation history
- decisions/outcomes
- integrity information
- required configuration excluding unsafe credential transfer

Target:

> **K6 dies → new machine**
> → install LEE
> → Restore Existing LEE
> → restore Brain
> → restore archive
> → reauthorize external accounts
> → verify systems
> → LEE continues.

Years of experience should not disappear with one computer.

47. MEMORY HEALTH

Create Memory Health as part of LEE's system health.

Measure things like:

- canonical DB health
- Event Log continuity
- archive availability
- semantic-index freshness
- orphaned evidence
- provenance completeness
- unresolved duplicate sources
- backup freshness
- restore-test status
- disk capacity

- archive capacity
- memory corruption detection
- stale consolidations
- excessive unresolved contradictions
- index rebuild requirements

Normally healthy memory should remain quiet.

Problems should appear prominently when they threaten cognition or durability.

48. MEMORY PRESSURE MANAGEMENT

If storage approaches capacity, LEE must not simply delete memories.

Use a staged pressure response:

21. Remove temporary artifacts.
22. Delete rebuildable caches.
23. Deduplicate.
24. Compress cold content.
25. Move archival material to colder storage.
26. Alert Jesse.
27. Recommend capacity expansion.

Canonical history should be the last thing considered for removal, and any destructive retention action should follow explicit policy and governance.

49. RETRIEVAL SHOULD BE MULTI-STAGE

LEE should not depend on semantic search alone.

A sophisticated retrieval process may combine:

- exact structured query
- entity graph traversal
- temporal filtering
- project relationships
- person relationships
- semantic similarity
- importance
- freshness
- provenance
- objective relevance
- previous usage
- contradiction state

Then rank candidates into the bounded Context Packet.

This gives LEE better memory than “vector search over everything.”

50. COGNITIVE MEMORY SHOULD BE EXPLAINABLE

Whenever a memory materially influences a consequential recommendation, LEE should be capable of answering:

- What memory did you use?
- Where did it come from?
- Is it a fact or interpretation?
- How old is it?
- Has it been contradicted?
- Why was it relevant?
- When was it last validated?
- What would change your conclusion?

That is critical for advanced cognition.

51. MEMORY AND CIL SHOULD WORK TOGETHER

LEE's memory system determines what knowledge may be relevant.

CIL determines how cognition should resolve the request and whether existing cognitive assets can avoid another frontier-model call.

Conceptually:

LEE Brain → Query Engine
→ Context Economy
→ relevant memory
→ CIL
→ reuse or model route
→ reasoning
→ structured result
→ potential new memory

New reasoning must not immediately become durable truth.

It should enter the appropriate category:

- candidate interpretation
- prediction
- proposed decision
- lesson candidate
- investigation update

and only become stronger knowledge through evidence and governance rules.

52. NIGHTLY COGNITIVE CONSOLIDATION

The earlier reflection concept should now become a serious subsystem.

A scheduled consolidation cycle could perform:

Observe

What meaningfully changed?

Reconcile

What new evidence conflicts with existing understanding?

Review

Which assumptions became stale?

Evaluate

Which predictions resolved?

Learn

Which decisions produced observable outcomes?

Consolidate

Which repeated experiences form lessons?

Institutionalize

Which patterns meet promotion requirements?

Compress

Which historical material can be represented more efficiently?

Archive

Which sources can move from hot to cold storage?

Curiosity

Which knowledge gaps now deserve investigation?

Prioritize

What deserves attention tomorrow?

Prepare

Build tomorrow's briefing.

The process should be observable and auditable.

LEE should never secretly rewrite the past during consolidation.

53. COGNITIVE CONTINUITY

LEE should eventually experience operational continuity across:

- conversations
- restarts
- model changes
- application updates
- Replit outages
- desktop migration
- K6 replacement
- connector outages

The model generating the current answer may change.

LEE's identity and accumulated experience should not.

The architecture should therefore treat the current LLM as a cognitive resource, not the holder of LEE's identity.

LEE's continuity comes from:

- Brain
- Event Log
- owner model
- self model
- experience
- institutional knowledge
- goals
- relationships
- decisions
- history
- constitutional boundaries

54. ADVANCED FUNCTIONAL AWARENESS

With the 20 original cognitive systems plus the memory architecture above, LEE should eventually maintain continuously updated versions of:

WORLD MODEL

What is happening outside LEE?

OWNER MODEL

What does Jesse value, want, know, prefer and need?

LAB MODEL

What is happening across Lamont Labs?

PROJECT MODEL

What is happening in each project?

RELATIONSHIP MODEL

Who matters and what is happening between people?

TEMPORAL MODEL

What happened, what changed and what is approaching?

SELF MODEL

What can LEE do, what is she doing and where are her limitations?

UNCERTAINTY MODEL

What does LEE not know?

GOAL MODEL

What outcomes are being pursued?

AUTHORITY MODEL

What may LEE do?

ATTENTION MODEL

What matters right now?

EXPERIENCE MODEL

What has worked and failed before?

MEMORY MODEL

What should be active, archived, consolidated, questioned or preserved?

These models should influence one another without collapsing into one undifferentiated record.

55. ADVANCED SELF-IMPROVEMENT BOUNDARY

LEE may evaluate:

- recommendation usefulness
- alert usefulness
- prediction accuracy
- retrieval effectiveness
- project diagnosis accuracy
- false positives
- ignored recommendations
- unnecessary context
- model spend
- CIL efficiency
- repair success
- owner corrections

LEE may propose:

- retrieval changes

- ranking changes
- attention thresholds
- consolidation improvements
- interface changes
- project-operation improvements
- new lessons
- candidate institutional knowledge

LEE must not have unrestricted authority to rewrite:

- Constitution
- CerbaSeal boundaries
- CIL authority
- owner identity
- security rules
- audit rules
- memory-integrity rules
- consequential-action permissions

The smarter LEE becomes, the more important that separation becomes.

56. MEMORY CONSTITUTION

I would add several permanent rules.

RULE 1 — HISTORY IS NOT REWRITTEN TO MATCH CURRENT BELIEF

LEE may supersede an interpretation, but must preserve that the previous interpretation existed.

RULE 2 — MODEL OUTPUT IS NOT AUTOMATICALLY FACT

Reasoning creates interpretations unless stronger evidence rules apply.

RULE 3 — PROVENANCE SURVIVES CONSOLIDATION

A lesson or institutional memory must remain traceable to its supporting experiences.

RULE 4 — ARCHIVAL IS NOT DELETION

Moving information to colder storage must not make LEE falsely claim the history never existed.

RULE 5 — INDEXES ARE REBUILDABLE

Semantic indexes and caches are retrieval aids, not canonical truth.

RULE 6 — OWNER TRUTH AND INFERENCE REMAIN DISTINCT

LEE must distinguish Jesse's explicit declarations from patterns inferred about Jesse.

RULE 7 — UNCERTAINTY IS VALID MEMORY

LEE may record “unknown” rather than invent an answer.

RULE 8 — OLD DOES NOT MEAN UNIMPORTANT

Age alone must never invalidate Strategic Anchors, constitutional rules or durable knowledge.

RULE 9 — STORAGE PRESSURE MUST NOT CAUSE SILENT FORGETTING

Capacity problems should trigger storage-management procedures and owner notification.

RULE 10 — THE BRAIN MUST OUTLIVE THE MODEL AND HARDWARE

LEE's continuity cannot depend on one AI model, one Replit workspace, or one K6.

57. WHAT THIS MAKES POSSIBLE

After years of operation, LEE might have:

- millions of events
- hundreds of thousands of emails
- thousands of documents
- years of calendar history
- hundreds of projects
- thousands of decisions
- extensive relationship history
- thousands of model interactions
- years of predictions
- years of lessons
- a large institutional-knowledge base
- enormous source archives

And yet a question like:

Where do things stand with Olivia?

does not load millions of records.

LEE identifies:

Olivia → Line Axia
→ CerbaSeal
→ pilot initiative
→ current objective
→ relevant email history
→ commitments
→ waiting state
→ latest meaningful evidence
→ related decisions
→ current communication pattern

and perhaps retrieves only 20–50 highly relevant memory objects.

That becomes the active cognitive context.

Everything else remains available without becoming active.

58. THE END-STATE EXPERIENCE

You could eventually return to the K6 after several days and LEE could say:

Welcome back.

I processed 43 meaningful events while you were away. Most were routine.

Three changed active objectives.

Olivia replied regarding the CerbaSeal offering. That moves the pilot initiative from Waiting to Action Required.

QuantraCore experienced a build failure. I traced it to a dependency change, prepared a repair and verified the safe test path. The modification is waiting for your approval.

Greyline completed two publication packages. Both passed structural validation. One cover has an unresolved low-confidence typography issue.

I resolved 82% of eligible reasoning requests through CIL without new frontier execution.

During consolidation, one previous assumption about Project X became stale, two predictions resolved correctly, and one repeated project failure pattern qualified as a lesson candidate.

Operational confidence is 96%.

Memory health is good. The Brain backup was restore-tested last night. The semantic index is current. Archive storage is at 41%.

My largest current knowledge gap is Google Drive because synchronization has been unavailable since 4:17 AM.

The most important thing requiring you is Olivia's message.

That would appear extraordinarily aware because LEE would have genuine functional continuity across attention, memory, time, relationships, goals, systems, uncertainty and experience.

The central product objective would then be:

LEE should not attempt to remember everything equally. She should preserve everything worth preserving, understand what each memory means, consolidate experience without falsifying history, retrieve the right memory at the right time, and know when she does not know enough.

That is the memory architecture I would put underneath the full cognition design.

And I would now make this a major formal Project LEE development track, not a later storage optimization. The cognition you want and the memory architecture are really one system.